

KSA 1 - Assignment

Discretize Continuous attributes:

Age (Sorted): 10, 14, 18, 22, 27, 29, 31, 37,

39, 45, 56, 59, 68 - split into 4 life-stage bins:

Bin	Range	Record Count
Child	0-18	10, 14, 18 (3)
Young Adult	19-35	22, 27, 29, 31 (4)
Middle Age	36-55	37, 39, 45 (3)
Senior	56+	59, 59, 65 (3)

Justification: these are standard demographic high stage cutoffs, and they give a reasonably balance split (3-4-3-3) across 3 records

Income (sorted): 0, 0, 12, 20, 32, 32, 47, 54,

59, 65, 72, 75, 87 - split 3 bracket bins:

Bin	Range	Records
Low	0-30K	0, 0, 12, 20 (4)
Medium	31-60K	32, 32, 47, 54, 59 (5)
High	61K	65, 72, 75, 87 (4)

Justification: mirrors common low/middle/high income brackets and yields a balance 4-5-4 split.

Discretized Dataset:

Age	Graduated	Income	Gender	Smoke
0-18	No	low	M	Yes
19-35	Yes	medium	F	Yes
19-35	No	Medium	M	Yes
0-18	No	Low	F	No
19-35	Yes	High	F	No
36-55	No	Medium	F	No
56+	yes	High	M	No
36-55	yes	High	M	No
56+	No	Medium	M	No
36-55	No	High	M	Yes
56+	Yes	Medium	F	No
0-18	No	Low	M	Yes
36-55	Yes	Low	F	Yes

Step 2:

Consider $\begin{bmatrix} \text{Yes} \rightarrow + \\ \text{No} \rightarrow - \end{bmatrix}$

Entropy of the entire dataset

$S(+6, -7)$

$$\text{Entropy}(S) = -\frac{6}{13} \log_2\left(\frac{6}{13}\right) - \frac{7}{13} \log_2\left(\frac{7}{13}\right)$$

$$= 0.9958$$

Age(0-18, 19-35, 36-55,
56+)

$$\begin{aligned}\text{Entropy for } 0-18(+2, -1) &= -\frac{2}{3} \log_2\left(\frac{2}{3}\right) \\ &\quad - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ &= 0.918\end{aligned}$$

$$\begin{aligned}\text{Entropy for } 19-35(+2, -2) &= -\frac{2}{4} \log_2\left(\frac{2}{4}\right) \\ &\quad - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{Entropy for } 36-55(+1, -2) &= -\frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ &\quad - \frac{2}{3} \log_2\left(\frac{2}{3}\right) \\ &= 0.918\end{aligned}$$

$$\begin{aligned}\text{Entropy for } 56+(+1, -2) &= -\frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ &\quad - \frac{2}{3} \log_2\left(\frac{2}{3}\right) \\ &= 0.918\end{aligned}$$

$$\begin{aligned}\text{Information Gain(Age)} &= 0.995 - \frac{3}{13} \times 0.918 \\ &\quad - \frac{4}{13} \times 1 - \frac{3}{13} \times 0.918 - \frac{3}{13} \times 0.918 \\ &= 0.052\end{aligned}$$

Graduated (Yes, No)

$$\text{Entropy for Yes (+2, -4)} = -\frac{2}{6} \log_2\left(\frac{2}{6}\right) - \frac{4}{6} \log_2\left(\frac{4}{6}\right) \\ = 0.918$$

$$\text{Entropy for No (+4, -3)} = -\frac{4}{7} \log_2\left(\frac{4}{7}\right) - \frac{3}{7} \log_2\left(\frac{3}{7}\right) \\ = 0.985$$

$$\text{Information Gain} = 0.9958 - \frac{6}{13} \times 0.918 \\ - \frac{7}{13} \times 0.985 \\ = 0.442$$

Income (High, Medium, Low)

$$\text{Entropy for Low (+3, -1)} = -\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \\ = 0.811$$

$$\text{Entropy for medium (+2, -3)} = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \\ -\frac{3}{5} \log_2\left(\frac{3}{5}\right) \\ = 0.971$$

5

$$\begin{aligned} \text{Entropy for high } (+1, -3) &= -\frac{1}{4} \log_2\left(\frac{1}{4}\right) - \\ &\quad -\frac{3}{4} \log_2\left(\frac{3}{4}\right) \\ &= 0.811 \end{aligned}$$

$$\begin{aligned} \text{Information Gain (Income)} &= 0.996 - \frac{4}{13} \times 0.811 \\ &\quad - \frac{5}{13} \times 0.971 - \frac{4}{13} \times 0.811 \\ &= 0.123 \end{aligned}$$

$$\underline{\text{Gender (M, F)}}$$

$$\begin{aligned} \text{Entropy for M } (4, -3) &= -\frac{4}{7} \log_2\left(\frac{4}{7}\right) - \frac{3}{7} \log_2\left(\frac{3}{7}\right) \\ &= 0.985 \end{aligned}$$

$$\begin{aligned} \text{Entropy for F } (+2, -4) &= -\frac{2}{6} \log_2\left(\frac{2}{6}\right) \\ &\quad - \frac{4}{6} \log_2\left(\frac{4}{6}\right) \\ &= 0.918 \end{aligned}$$

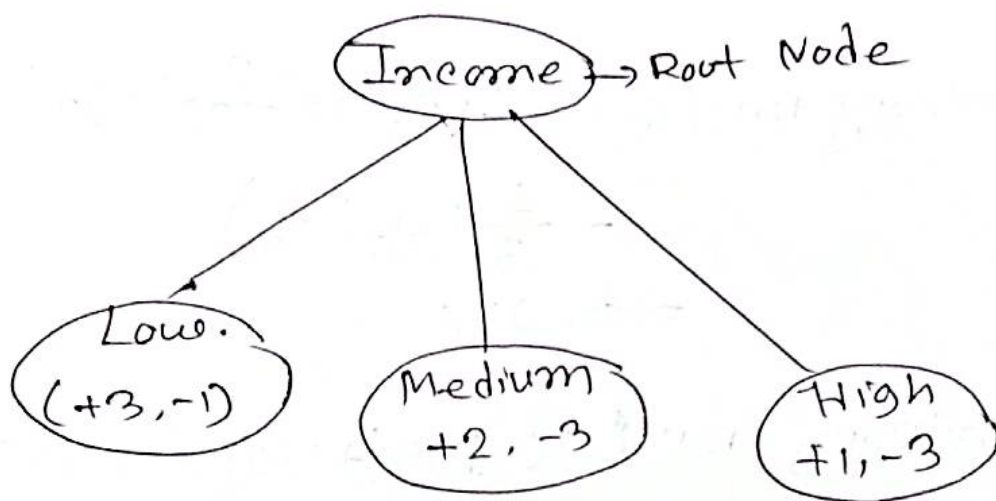
$$\begin{aligned} \text{I. Gain (Gender)} &= 0.996 - \frac{7}{13} \times 0.985 - \frac{6}{13} \times 0.918 \\ &= 0.042 \end{aligned}$$

$$\text{Gain}(S, \text{age}) = 0.052$$

$$\text{Gain}(S, \text{Graduated}) = 0.042$$

$$\text{Gain}(S, \text{Income}) = 0.123 \text{ (highest)}$$

$$\text{Gain}(S, \text{Gender}) = 0.042$$



Income Low:

age	Graduated	Income	Gender	Smoke
0-18	NO	Low	M	Yes
0-18	NO	Low	F	No
0-18	NO	Low	M	yes
36-55	yes	Low	F	yes

$$\begin{aligned}\text{Entropy for Income Low}(+3, -1) &= -\frac{3}{4} \log_2\left(\frac{3}{4}\right) \\ &\quad - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \\ &= 0.811\end{aligned}$$

Age

$$\begin{aligned}\text{Entropy for } (0-18)(+2, -1) &= -\frac{2}{3} \log_2\left(\frac{2}{3}\right) \\ &\quad - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ &= 0.918\end{aligned}$$

$$\begin{aligned}\text{Entropy for } \{36-55\}(+1, 0) &= -\frac{1}{1} \log_2\left(\frac{1}{1}\right) \\ &\quad - \frac{0}{1} \log_2\left(\frac{0}{1}\right) \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{Information gain(Age)} &= 0.811 - \frac{3}{4} \times 0.918 \\ &= 0.123\end{aligned}$$

Graduation (No, Yes)

$$\begin{aligned}\text{Entropy for No (+2, 1)} &= -\frac{2}{3} \log_2\left(\frac{2}{3}\right) \\ &\quad - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ &= 0.918\end{aligned}$$

$$\text{Entropy for Yes (+1, 0)} = 0$$

$$\begin{aligned}\text{Information Gain (Graduation)} &= 0.811 - \frac{3}{4} \times 0.918 \\ &= 0.123\end{aligned}$$

Gender (M, L)

$$E_{\text{Male}}(+2, 0) = 0$$

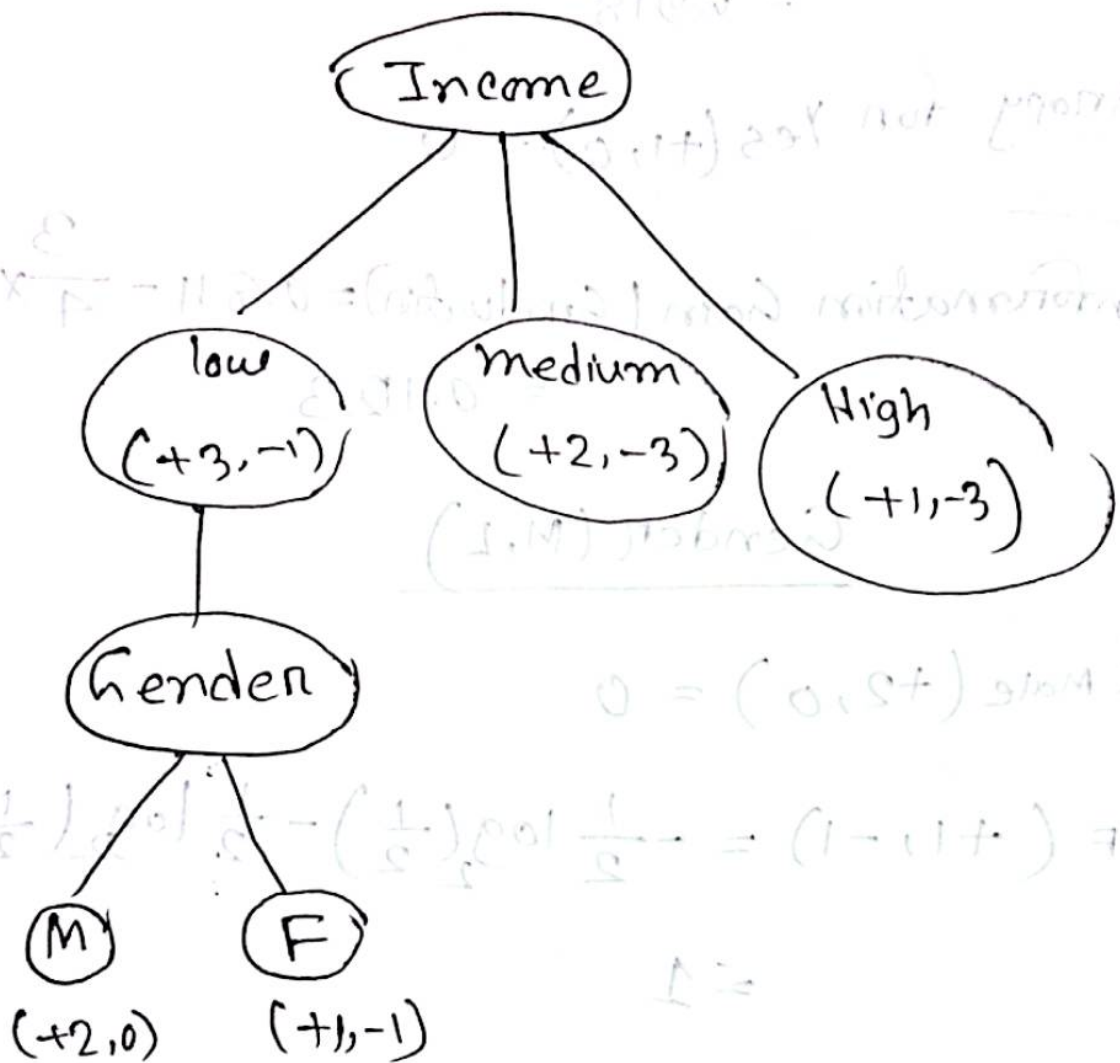
$$\begin{aligned}E_F(+1, -1) &= -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{I. Gain (Gender)} &= 0.811 - \frac{2}{4} \times 0 - \\ &\quad \frac{2}{4} \times 1 = 0.311\end{aligned}$$

$$\text{Gain}(S_{\text{low}}, \text{age}) = 0.918$$

$$\text{Gain}(S_{\text{low}}, \text{graduation}) = 0.123$$

$$\text{Gain}(S_{\text{low}}, \text{gender}) = 0.311 \rightarrow \text{highest}$$



Yes

Gender split:

Age	Graduation	Income	Gender	Smoke
0-18	No	Low	F	No
31-55	Yes	Low	F	Yes

Entropy for Gender(F) (+1, -1)

$$E = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right)$$

$$= 1$$

Age

$$E_{0-18}(0, +1, -1) = -\frac{0}{1} \log_2\left(\frac{0}{1}\right) - \frac{1}{1} \log_2\left(\frac{1}{1}\right)$$

$$= 0$$

$$E_{(31-55)}(+1, -0) = 0$$

$$I.G(Age) = 1 - 0$$

$$= 1$$

Graduation:

$$E_{No}(+0, -1) = 0$$

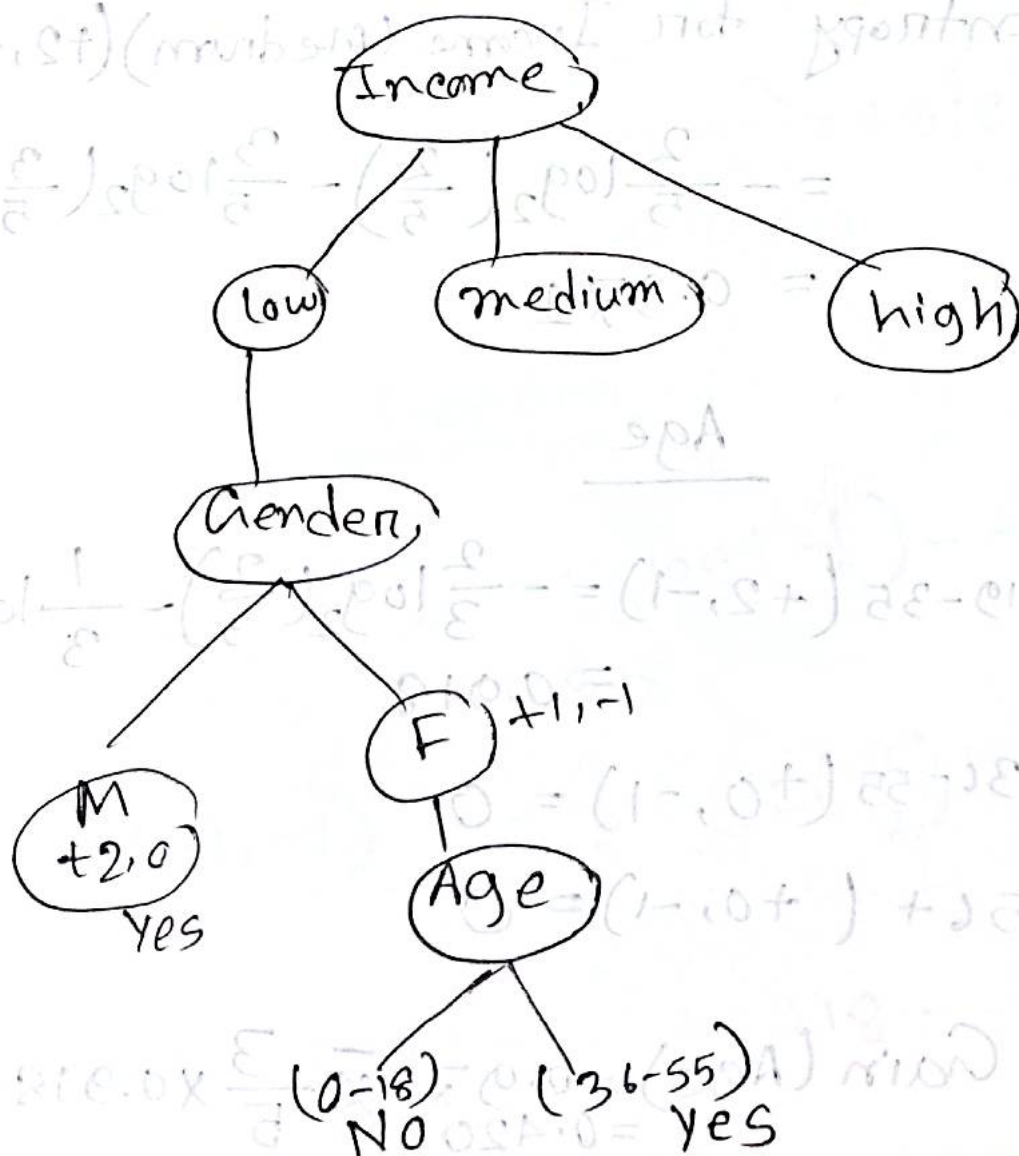
$$E_{Yes}(+1, 0) = 0$$

$$\text{Information gain (araduation)} = 1 - 0 - 0 = 1$$

$$\text{Gain}(S_F, \text{age}) = 1$$

$$\text{Gain}(S_F, \text{araduation}) = 1$$

both are same. So by left to right priority Pick.



Income medium:

Age	Graduated	Income	Gender	Smoke
19-35	Yes	medium	F	Yes
19-35	NO	"	M	Yes
36-55	NO	"	F	NO
19-35	NO	"	M	NO
56+	Yes	"	F	NO

Entropy for Income (medium) (+2, -3)

$$= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right)$$

$$= 0.971$$

Age

$$E_{19-35} (+2, -1) = -\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right)$$

$$= 0.918$$

$$E_{36-55} (+0, -1) = 0$$

$$E_{56+} (+0, -1) = 0$$

$$I. \text{ Gain (Age)} = 0.971 - \frac{3}{5} \times 0.918$$

$$= 0.420$$

Graduation

$$E_{yes}(+1, -1) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right)$$
$$= 1$$

$$E_{no}(+1, -2) = -\frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right)$$
$$= 0.918$$

$$I. Gain(Graduation) = 0.971 - \frac{2}{5} \times 1$$
$$= 0.020$$

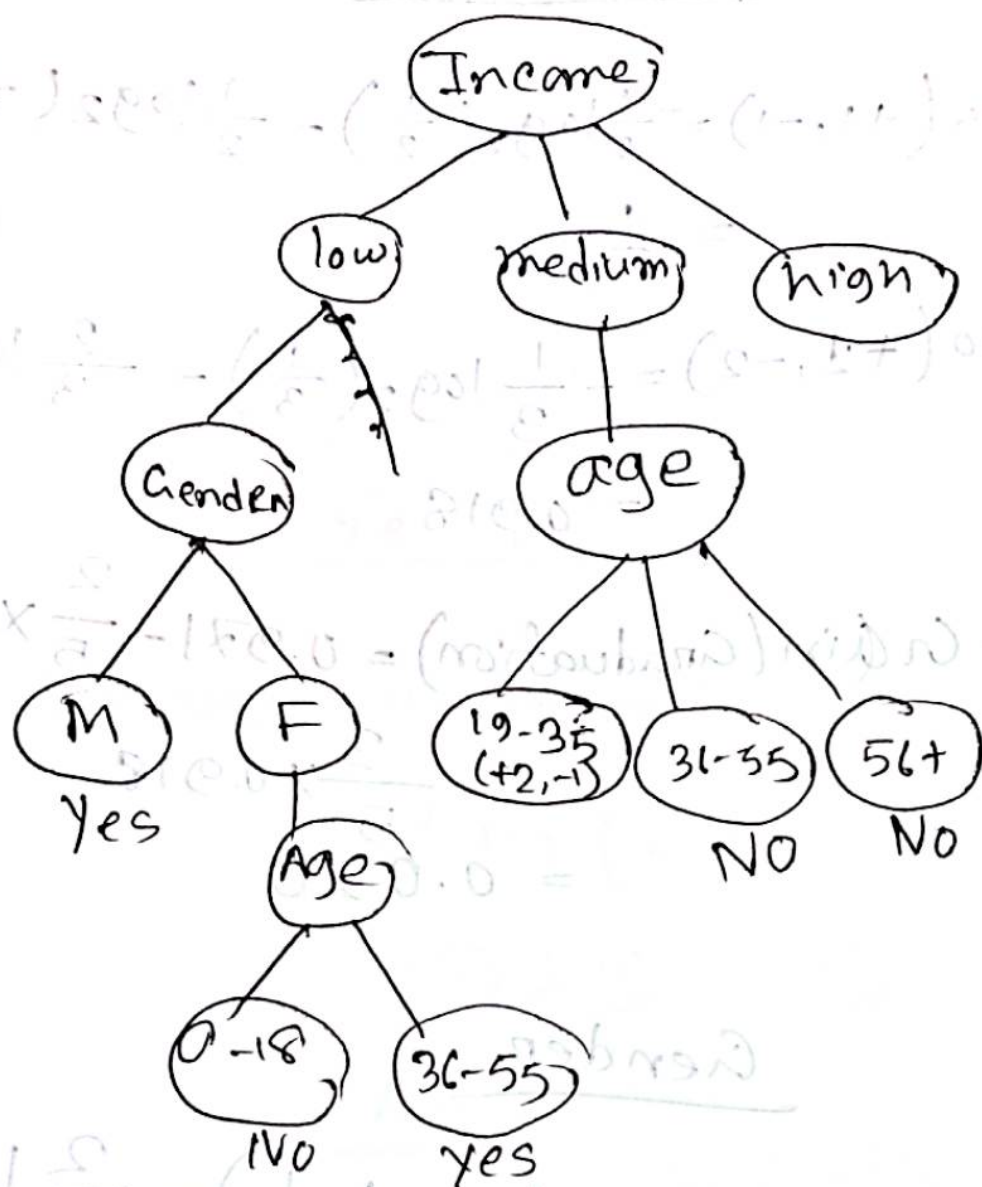
Gender

$$E_F(+1, -2) = -\frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right)$$
$$= 0.918$$

$$E_M(+1, -1) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right)$$
$$= 1$$

$$I. Gain = 0.971 - \frac{3}{5} \times 0.918 - \frac{2}{5} \times 1$$
$$= 0.020$$

I. Gam (Age) → highest.



Age splited:

$$\text{Entropy}(+2, -1) = -\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \\ = 0.918$$

Graduation:

$$E_{\text{yes}}(+1, 0) = 0$$

$$E_{\text{no}}(+1, -1) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \\ = 1$$

$$\text{I. Gain(Graduation)} = 0.918 - 0 - \frac{2}{3} \times 1 \\ = 0.251$$

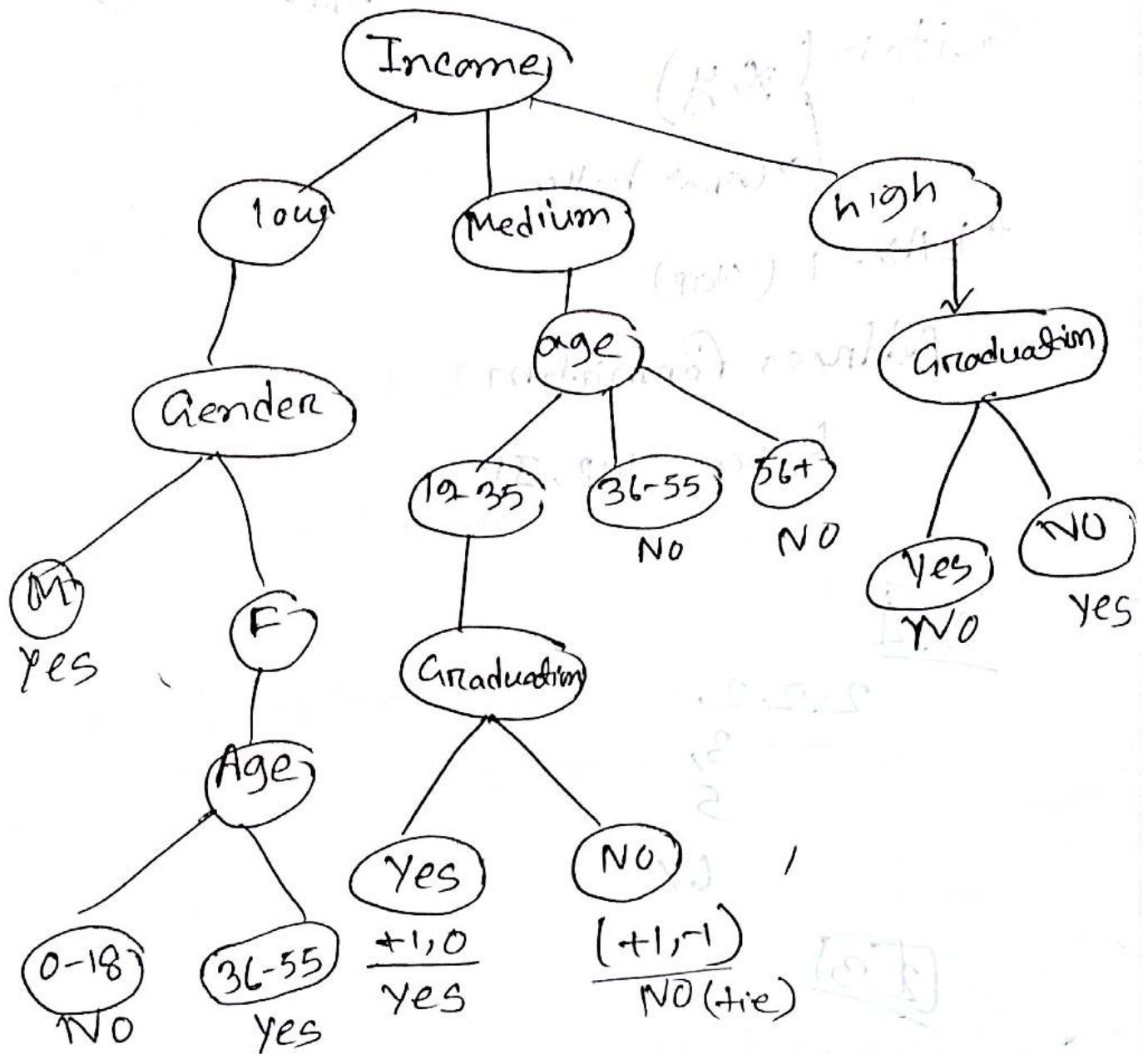
Gender

$$E_F(+1, 0) = 0$$

$$E_M(+1, -1) = 1$$

$$\text{I. Gain(Gender)} = 0.918 - \frac{1}{3} \times 0 - \\ \frac{2}{3} \times 1 \\ = 0.251$$

choose Left to right



Income High :

Age	Graduation	Income	Gender	Smoke
19-35	yes	High	F	NO
56+	yes	high	M	NO
36-55	yes	high	M	NO
56+	NO	high	M	Yes

$$\text{Entropy Income High } (+1, -3) = -\frac{1}{4} \log_2\left(\frac{1}{4}\right) - \frac{3}{4} \log_2\left(\frac{3}{4}\right)$$

$$= 0.811$$

Age

$$E_{19-35}(0, -1) = 0$$

$$E_{36-55}(0, -1) = 0$$

$$E_{56+}(+1, -1) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) = 1$$

$$\text{I. Gain (Age)} = 0.811 - \frac{2}{4} \times 1 = 0.311$$

Graduation

$$E_{\text{Yes}}(+0, -3) = 0$$

$$E_{\text{No}}(+1, 0) = 0$$

$$\text{I. Gain (Graduation)} = 0.811$$

Gender

$$E_F(+0, -1) = 0$$

$$E_M(+1, -2) = 0 - \frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right)$$
$$= 0.918$$

$$I. \text{ Gain (Gender)} = 0.811 - \frac{1}{3} \times 0 - \frac{2}{3} \times 0.918$$
$$= 0.123$$

Now, Graduation = Split:

S. Yes (+0, -3)

S. No (+1, 0)

